

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fifth Semester of B. Tech. Examination (EC)

December 2011

EC304 Power Electronics

Date: 15.12.2011, Thursday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

- Q - 1 (a) Explain the operation of SCR using two transistor analogy by deriving required equations. [04]
- (b) Explain the function of free wheeling diode with the inductive load in controlled rectifiers. [03]
- Q - 2 (a) Explain the reverse recovery characteristics of diode. [04]
- (b) Explain the series connections of SCR and discuss how to overcome problems related with it. [04]
- (c) What is cycloconverter? Explain the principal of operation of single phase to single phase cycloconverter. Sketch the output voltage waveforms for resistive load. [06]

OR

- Q - 2 (a) Write short note on Snubber Circuits [04]
- (b) An SCR with latching current of 4mA is connected in series with inductance of 0.1H with DC source voltage 100V. Obtain the minimum width of the gating pulse required to properly turn on the SCR. [04]
- (c) Explain working of single phase AC voltage bidirectional controller with resistive loads. [06]
- Q - 3 (a) The reverse recovery time of a diode is $t_{rr} = 3 \mu s$ and the rate of fall of the diode current is $di/dt = 30 A/\mu s$. Determine (i) the storage charge Q_{RR} , and (ii) the peak reverse current I_{RR} . [03]
- (b) List different types of commutation techniques and explain complementary commutation. [03]
- (c) Draw the circuit and output waveforms for 3- Φ semi converter with highly inductive load for firing angle (α) of 30° . Also derive the equations for output DC and RMS voltages for $\alpha \leq \pi/3$. [08]

OR

- Q - 3 (a) Explain regenerative mode and dynamic braking mode of DC motor. [03]
(b) Explain basic series inverter. [03]
(c) Explain single phase full converter for highly inductive load with the help of circuit diagram. Draw supply voltage, output voltage, and supply current wave form. Derive equation for average output voltage and RMS value of output voltage. [08]

SECTION – II

- Q - 4(a) Discuss the following with respect to BJT [03]
i) Secondary breakdown ii) Safe operating area
(b) Discuss the performance parameters of inverters. [04]
- Q - 5 (a) Draw and explain the structure of power MOSFET. Also explain principal of operation and switching characteristics of power MOSFET. [08]
(b) Explain the microcomputer control of DC drive. [06]

OR

- Q - 5 (a) Explain the operation of 3-phase bridge inverter operating in 180° mode. Draw appropriate waveforms. [08]
(b) What is the basis for the classification of DC converters? Indicate with necessary operating regions for all types. Explain two-quadrant converter (class C chopper) with circuit diagram. [06]
- Q - 6 (a) Draw the circuit of buck regulator and explain its working with equivalent circuit and waveforms. Obtain equation for ΔI and ΔV_c . [08]
(b) Draw and explain Full wave controlled rectifier with the inductive load with suitable waveforms and equations. [06]

OR

- Q - 6 (a) Explain the principal of step up chopper. Discuss the modes of operation of the circuit with necessary equivalent circuits and waveforms. Establish the condition for power controllable transfer. [08]
(b) Compare half bridge and full bridge inverter with circuit and waveforms for each. [06]
